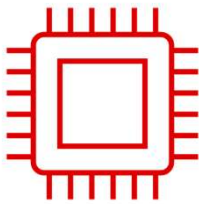


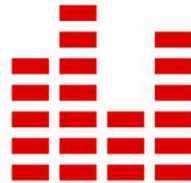
Design Of Differential-mode (DM) Active EMI Filter for ISOP LLC converters

Suhani Biswal

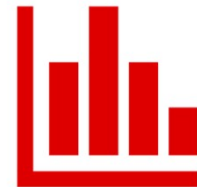
Objectives of This Work



*Design and simulation of a
DM Active EMI filter*



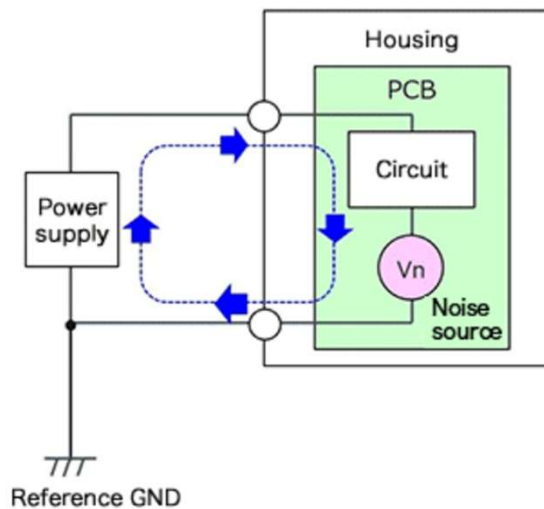
*Studying DM noise
spectrums at the LISN under
various conditions with and
without AEF*



*Ensuring EMC compliance is
achieved in the Sentry
schematic according to
CISPR 32 standards*

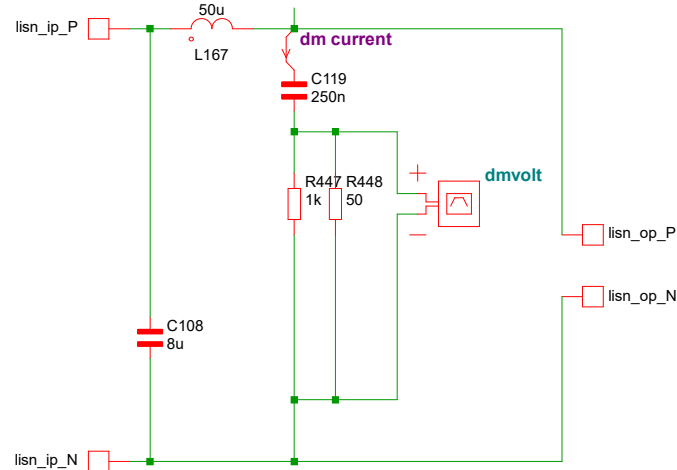
Differential-mode (DM) noise | Introduction

Differential (normal) mode noise

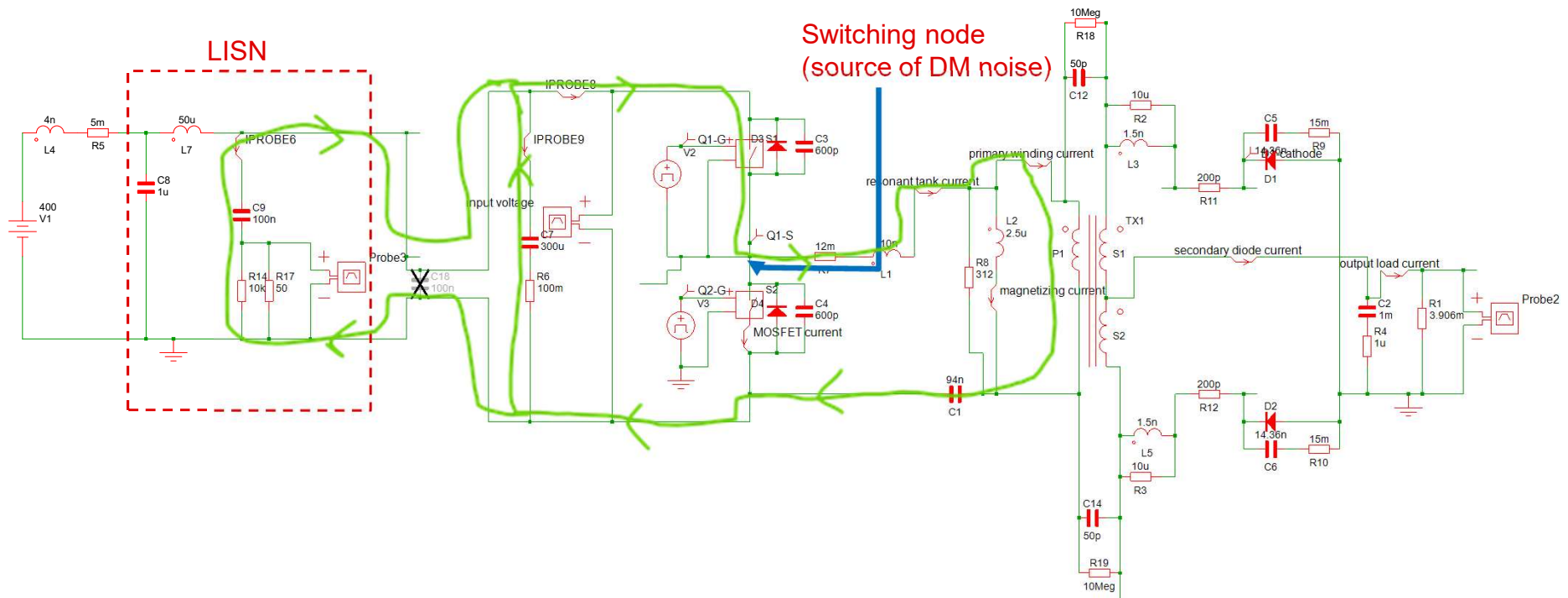


- A mode in which a noise current flows on the same path as the power supply current
- Noise voltage occurs across power supply lines

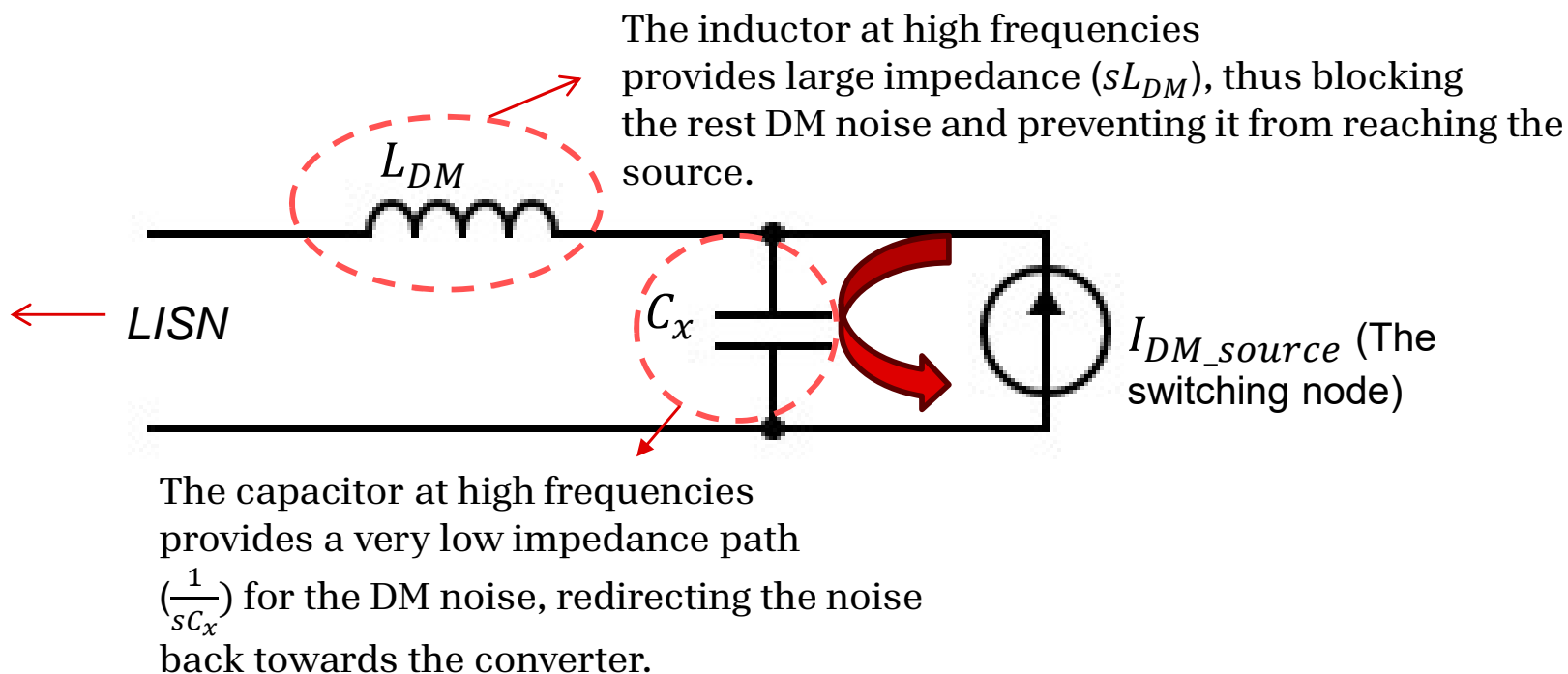
- In differential mode noise, the noise source appears across power supply lines and is in series with the power supply line, and the noise current flows in the same direction as the power supply current. It is called “differential mode” because **the outgoing and return currents are oppositely-directed**.
- The source of differential mode noise is the switching node, which causes high di/dt due to coupling with nearby capacitors, that travels across the circuit and reaches the power lines.
- For DM noise a single unit of LISN is required, connected between the line and neutral rails. For this project I’ve used the **50uH/8uF LISN setup**, which is used in industrial or residential applications. It mimics the **SIMPLIS LISN module** available for the same specifications.



DM noise path in LLC converter circuit (With LISN)



Passive filters for suppressing DM noise



Active EMI Filters | Introduction

The Need

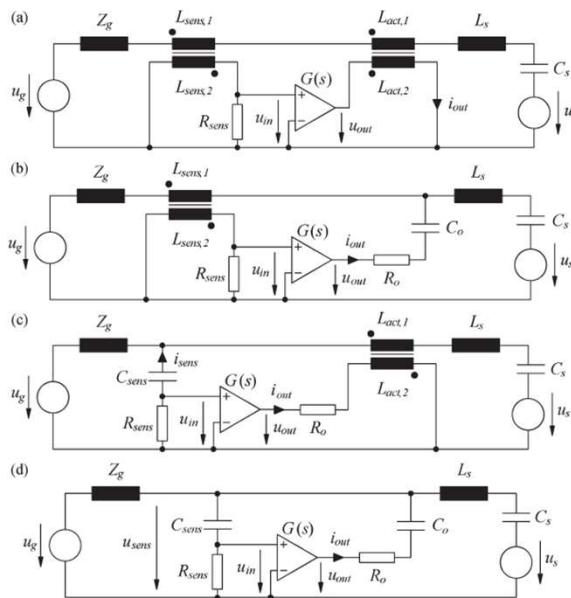
DM inductors dominate the size of the EMI filter, and bring several challenges during EMI filter design – including bulky volume, thermal management issues, acoustic noise, filter resonances and electromagnetic coupling between components.

In addition, the X-capacitors used for passive filtering in high voltage applications are also very bulky and consume a lot of space

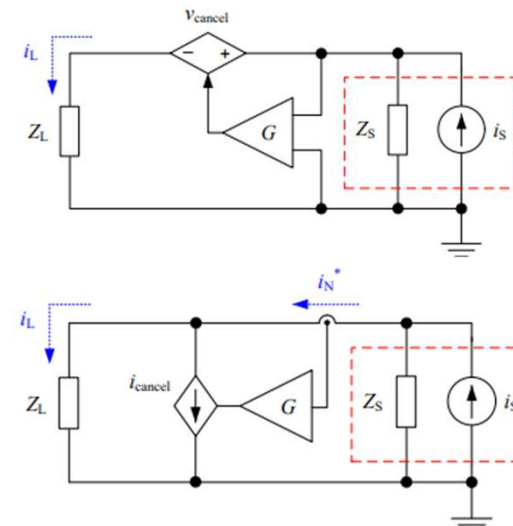
- The AEF circuit connects to the lines between the EMI source and the EMI victim.
- Unlike a passive filter, an AEF circuit uses active devices and control **to sense the residual (DM or CM) voltage or current disturbance, and injects an opposing signal that directly negates that noise disturbance**, essentially a destructive interference.
- The expectation is that AEF significantly reduces EMI, resulting in a smaller-size filter versus a traditional passive-only design with equivalent attenuation.

Active EMI Filters | Configurations

- The design and implementation of AEF and HEF circuits depend on the path of conduction (DM or CM) and the required **sensing**, **gain** and **injection** stages.
- There are two kinds of control structures: Feedback (FB) and Feedforward (FF)



Feedback topologies a) CSVI, b) CSCI, c) VSVI, d) VSCI



Feedforward topologies VSVI and CSCI

Selection of Active Filter Topology: Why Avoid Current Transformers?

Wide bandwidth requirement

EMI suppression must be effective from 150 kHz to 30 MHz, requiring a current transformer with excellent high-frequency response.

Magnetic material limitations

Conventional core materials exhibit reduced permeability at high frequencies, degrading sensing accuracy.

Self-resonance issues

Transformer parasitics can create unwanted resonances that affect filter stability and attenuation.

Increased size and cost:

High-performance HF current transformers add significant volume and expense to the EMI filter.

Thus, our choice :

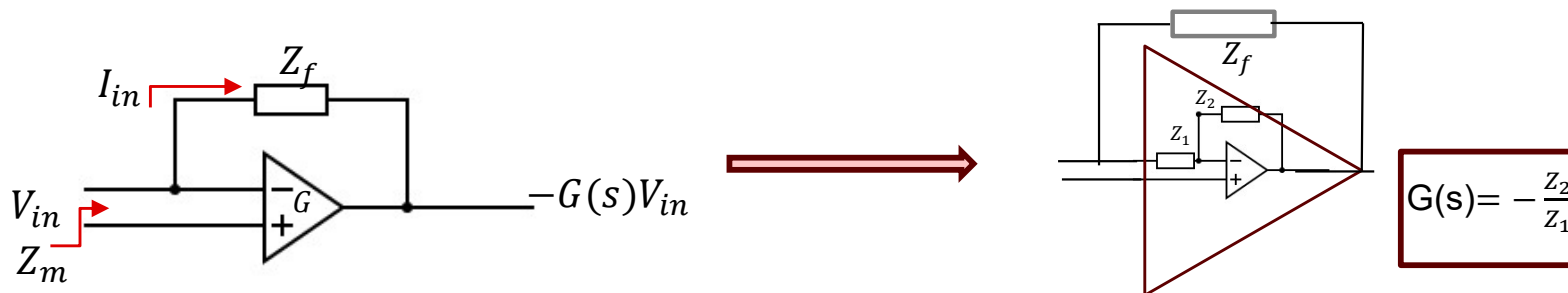
A **Voltage-Sensing Current-Injection (VSCI)** topology using capacitive sensing and capacitive injection was selected.

BENEFITS:

- *Eliminates bulky HF current transformers.*
- *Smaller size and lower cost.*
- *Better suitability for high-frequency EMI suppression.*
- *Easier integration with existing X-capacitor-based differential-mode filters.*

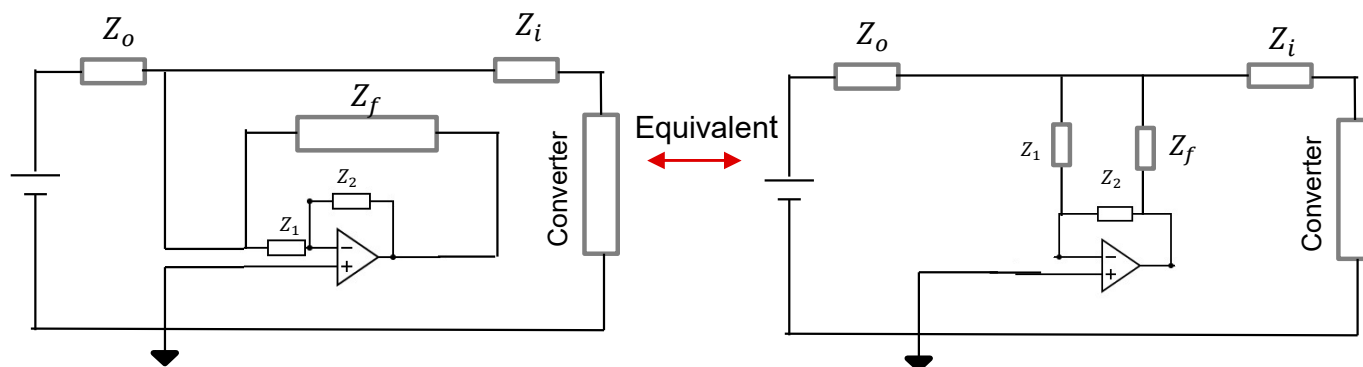
Capacitive Amplification

- The basic principle of the topology is to increase the equivalent capacitance of the x-capacitor for high frequencies by using a feedback loop. This is done using the **Miller effect**.

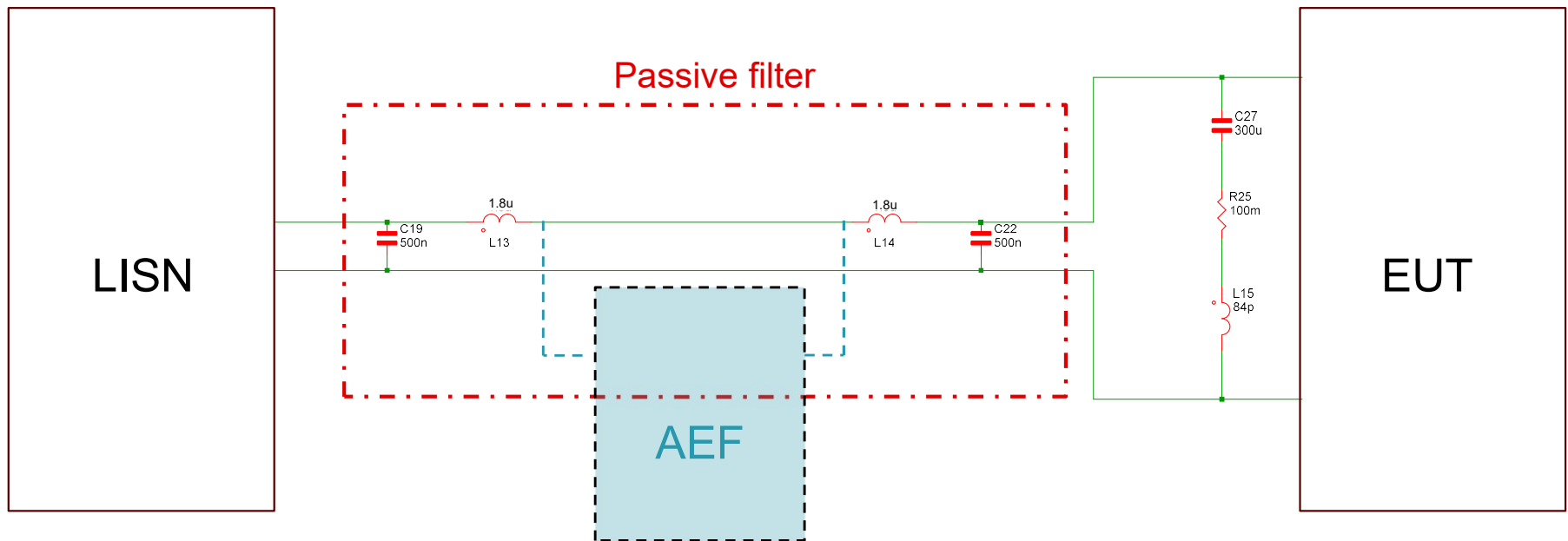


$$I_{in} = \frac{V_{in}(1+G(s))}{Z_f} \Rightarrow \frac{V_{in}}{I_{in}} = Z_m = \frac{Z_f}{(1+G(s))} \Rightarrow C_m = C_f(1 + G(s)) \uparrow$$

Thus the overall setup of an AEF is something like this:



Choice of Passives



The cut-off frequency is given by: $f_c = \frac{1}{2\pi\sqrt{LC}} = 118.63\text{kHz}$

where L is $2 \times 1.8\mu\text{H} = 3.6\mu\text{H}$ and C is 500nF. Thus the attenuation is already at a roll-off stage when we enter the EMI range at 150kHz.

Choice of Passives| Topology

Choice : π – LC filter | Question arises : Why not a simple LC filter?

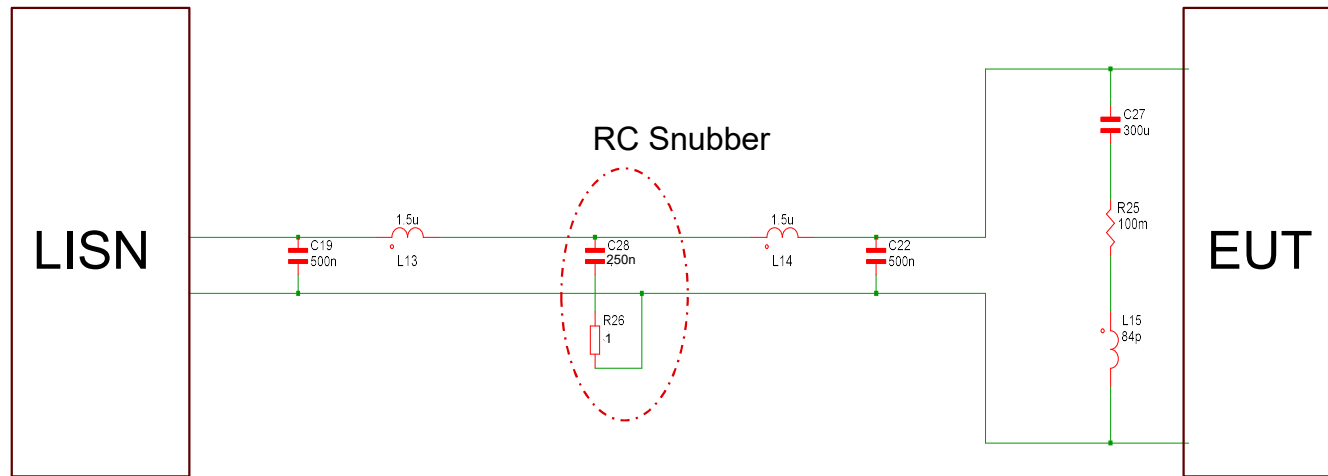
Reason : The op-amp circuitry is very sensitive to high current peaks because it saturates easily and adds to the high frequency peaks itself

(i) **Isolation from both sides** (*Protected bubble where the impedance is manageable*)

- a) From EUT side:** The high di/dt spikes from the switching components are converted into high frequency dv/dt due to the bulk capacitor's ESR and ESL. These high frequency spikes are absorbed by the 500nF x-capacitor. The 1.5uH inductor isolates the C_{inj} capacitor from the x-capacitor, which would otherwise be almost short circuited at frequencies above 500kHz. Thus, pushing heavy current through the C_{inj} , instantly saturating the op-amp.
- b) From LISN side :** The LC-pair at the LISN side helps localizing the cancellation current while maintaining a controlled impedance region. The LC wall also protects the op-amp from any source side EMI from other devices creeping in and corrupting the op-amp.

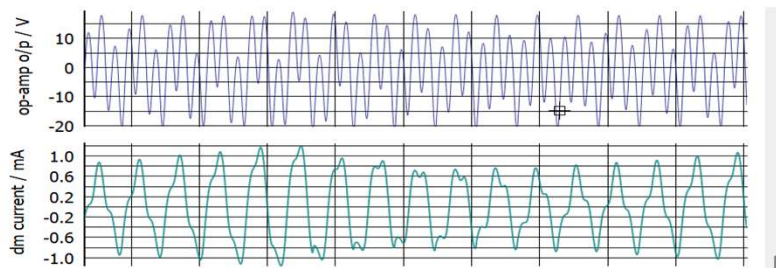
(ii) **Better attenuation :** π – LC filter gives -60dB/decade attenuation compared to simple LC filter that gives -40dB/decade. The cut-off frequency is 118kHz . Thus it crushes the noise from 150kHz to 30 MHz with a steeper roll-off.

Choice of Passives | Addition of snubber



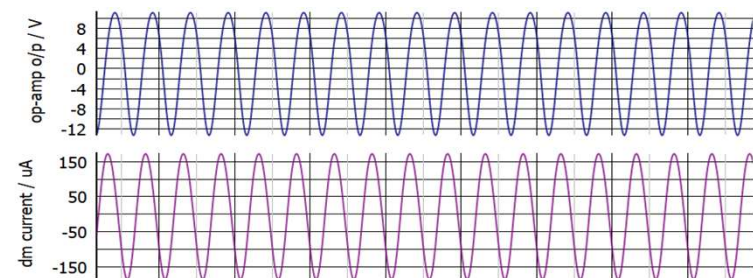
A snubber circuit is added to dampen LC resonance occurring in the passives. Without the snubber, a large low frequency component is introduced to the op-amp output. It distorts the waveform and increases the dm current.

Without snubber

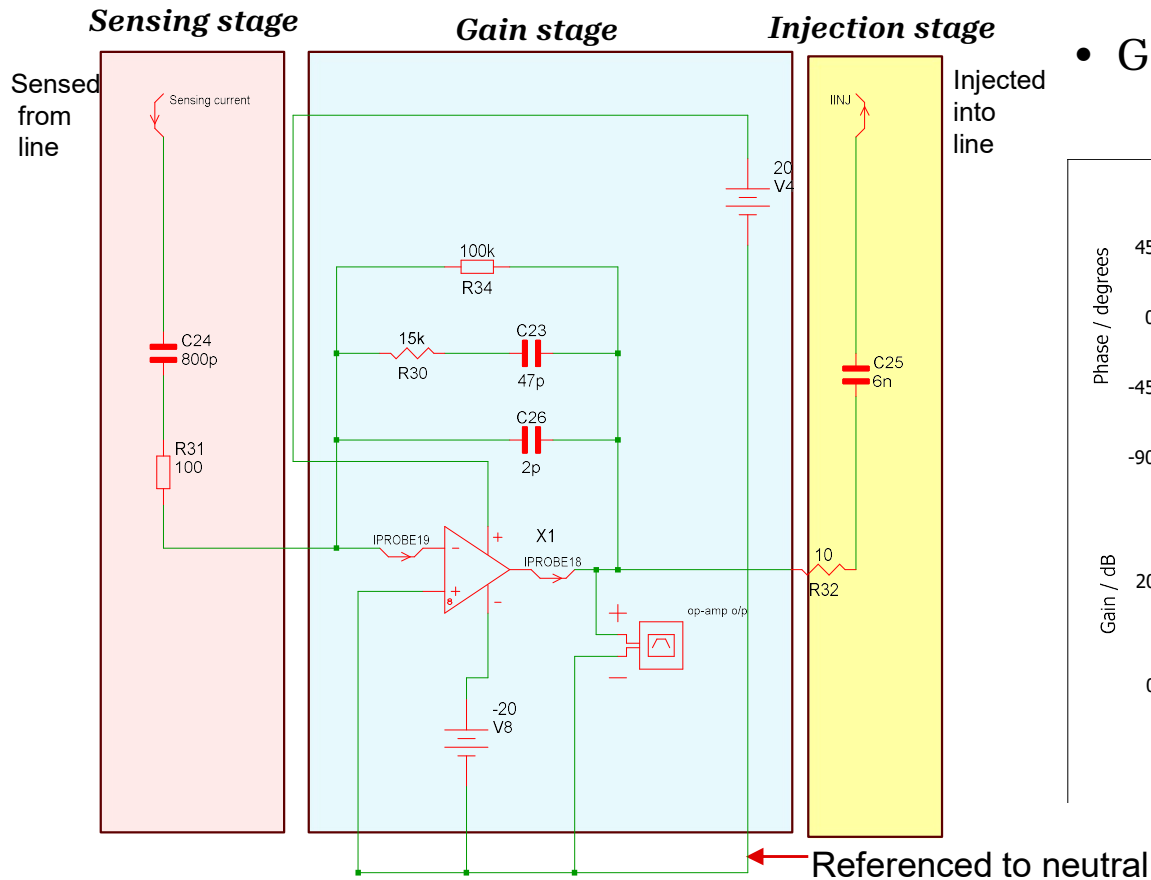


TI Information – Selective Disclosure

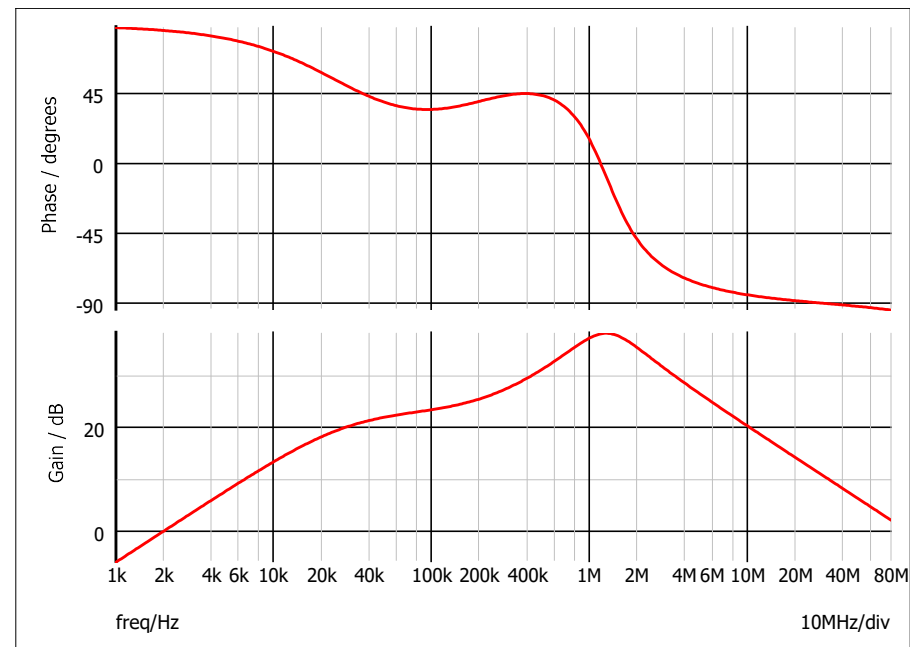
With snubber



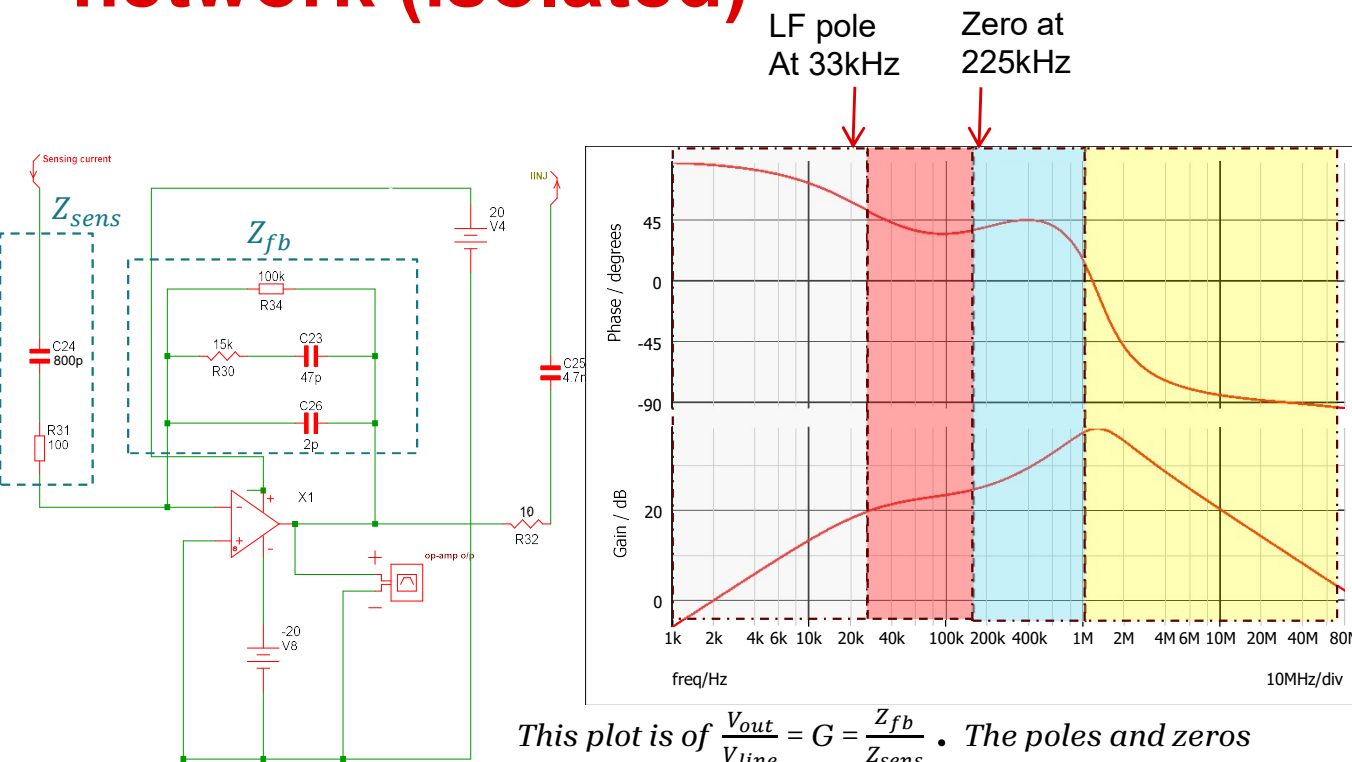
Components of proposed DM EMI filter



- Gain and Phase Plot of the standalone AEF:



Gain and Phase curves of the compensation network (isolated)



This plot is of $\frac{V_{out}}{V_{line}} = G = \frac{Z_{fb}}{Z_{sens}}$. The poles and zeros introduced by the compensation networks are as follows:

$$f_{p1} = \frac{1}{2\pi \cdot R34 \cdot C23} = 33\text{kHz} \quad f_z = \frac{1}{2\pi \cdot R30 \cdot C23} = 225\text{kHz}$$

$$f_{p2} = \frac{1}{2\pi \cdot R30 \cdot C2} = 5.3\text{MHz}$$

1. The Differentiator Climb: The $R34$ in the compensation network dominates and the gain = $sC24 \cdot R34$. Thus opamp acts as a differentiator. Hence gain slope is +20dB/dec and phase starts from +90 degrees.

2. The Baseline gain region : The first pole is at 33kHz. It adds a -20dB/dec. flattening the zone. The gain here is given by $C24/C23 = 24.6\text{dB}$. It sets a baseline gain before the zero hits, thus selection of these parameters is incredibly important. The phase plummets to around 40 degrees .

3. The Stabilizing zero: The zero is at 225kHz. It adds a +20dB/dec rise in slope allowing the opamp to fight harder in the critical 500kHz to 1MHz zone. The phase stops dropping here and is held steady.

4. The silicon limit is reached: The gain hits a peak of roughly 40dB around 1MHz and then abruptly rolls over and starts falling. Mathematically the HF pole at 6MHz should have held the gain stable at around 55dB. But it is greater than maximum possible gain at 5.3MHz that is around 25dB. This is due to the Gain Bandwidth Product (GBW) of the opamp. The gain slope gradually rolls off at -20dB/ dec.

Control loop block diagram of the AEF

The reference is kept as 0 Amp as the target sensed noise current is 0 Amp.

$$\frac{V_{out}}{V_{line}} = G = - \frac{Z_{fb}}{Z_{sens}}$$

$$Y_{sens} = \frac{1}{R_{sen} - jX_{c_{sen}}}$$

$$I_{inj} = \frac{V_{out} - V_{line}}{R_{inj} + \frac{1}{sC_{inj}}} = \frac{V_{line}(G-1)}{R_{inj} + \frac{1}{sC_{inj}}}$$

$$Y = \frac{I_{inj}}{V_{out}} = \frac{V_{line}(G-1)}{GV_{line}(R_{inj} + \frac{1}{sC_{inj}})}$$

$$Y = \frac{(G-1)}{G(R_{inj} + \frac{1}{sC_{inj}})}$$

$$V_{line} = Z_{eq}(s) \times (I_{inj} + I_{dm_source})$$

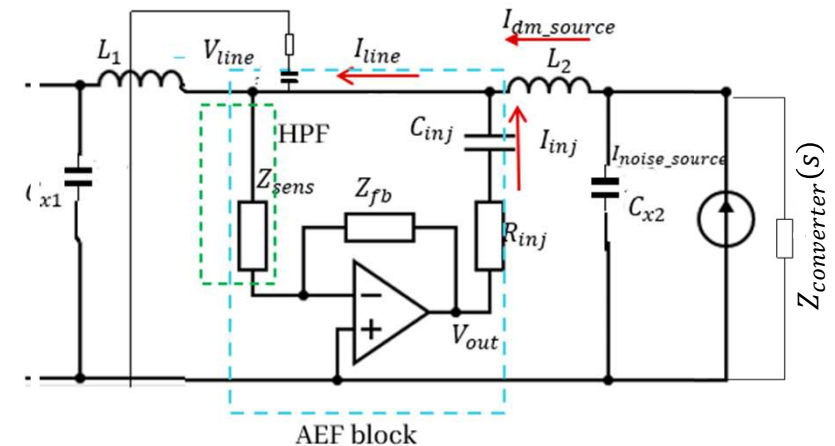
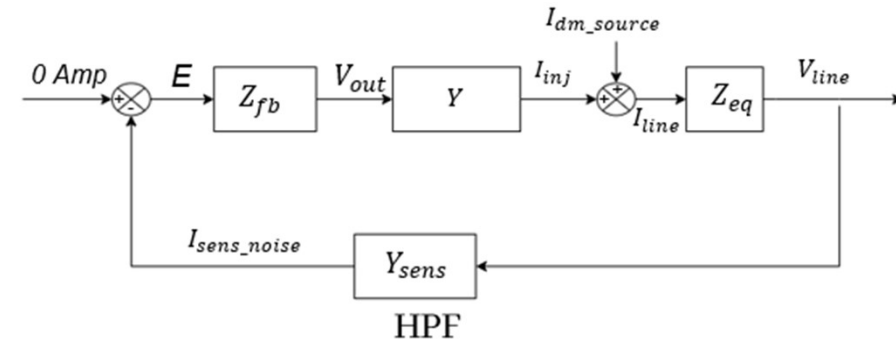
Where I_{dm_source} is the DM noise generated at the switching node

At the injection node, the current splits into four parallel paths, $Z_{eq}(s)$ is the **parallel combination** of these physical branches :

- Source Path: $Z_{source}(s) = sL_1 + \frac{1}{sC_{x1}}$
- Load or EUT Path : $Z_{load}(s) = sL_2 + (\frac{1}{sC_{x2}} \parallel Z_{converter}(s))$
- Snubber Path : $Z_{snub}(s) = R_d - jX_{C_d}$

($Z(s)$ includes resonant tank parameters and the load reflected to the primary side)
Thus,

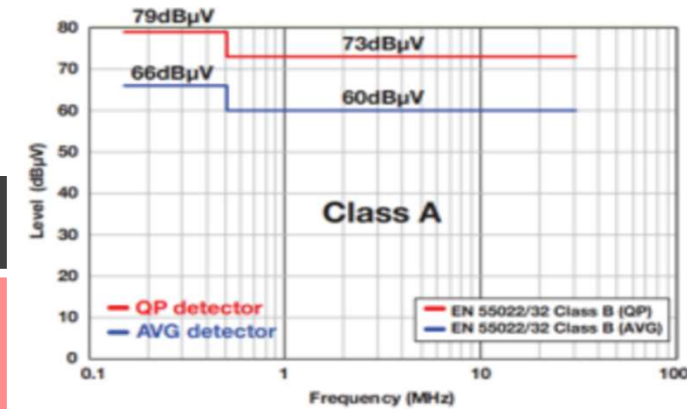
$$Z_{eq}(s) = Z_{source}(s) \parallel Z_{load}(s) \parallel Z_{snub}(s)$$



Results| Summary

- Target EMI limit : CISPR 32 Class A

Sl.no.	Condition	Highest peaks level (in dBuV)	Comments
1	No filter	136.32 (1MHz, 2MHz, 3MHz peaks, closely followed by rest HF peaks)	The whole spectrum from 150kHz to 30MHz was above 80dBuV (very high noise content).
2.	With π – LC filter (1.8uH and 500nH) with snubber circuit	74.40 (2MHz peak only)	Low noise, yet not adhering to standards yet (both QP and AVG detectors fail)
3.	With π – LC filter (1.8uH and 500nH) with snubber circuit + AEF	59.33 (2MHz peak only)	Low noise safely below the QP limit and just below AVG limit too.



The EMI voltage being measured is a continuous signal in time domain. Hence its peak = quasi-peak

Thus it passes both QP and AVG detectors with the AEF.

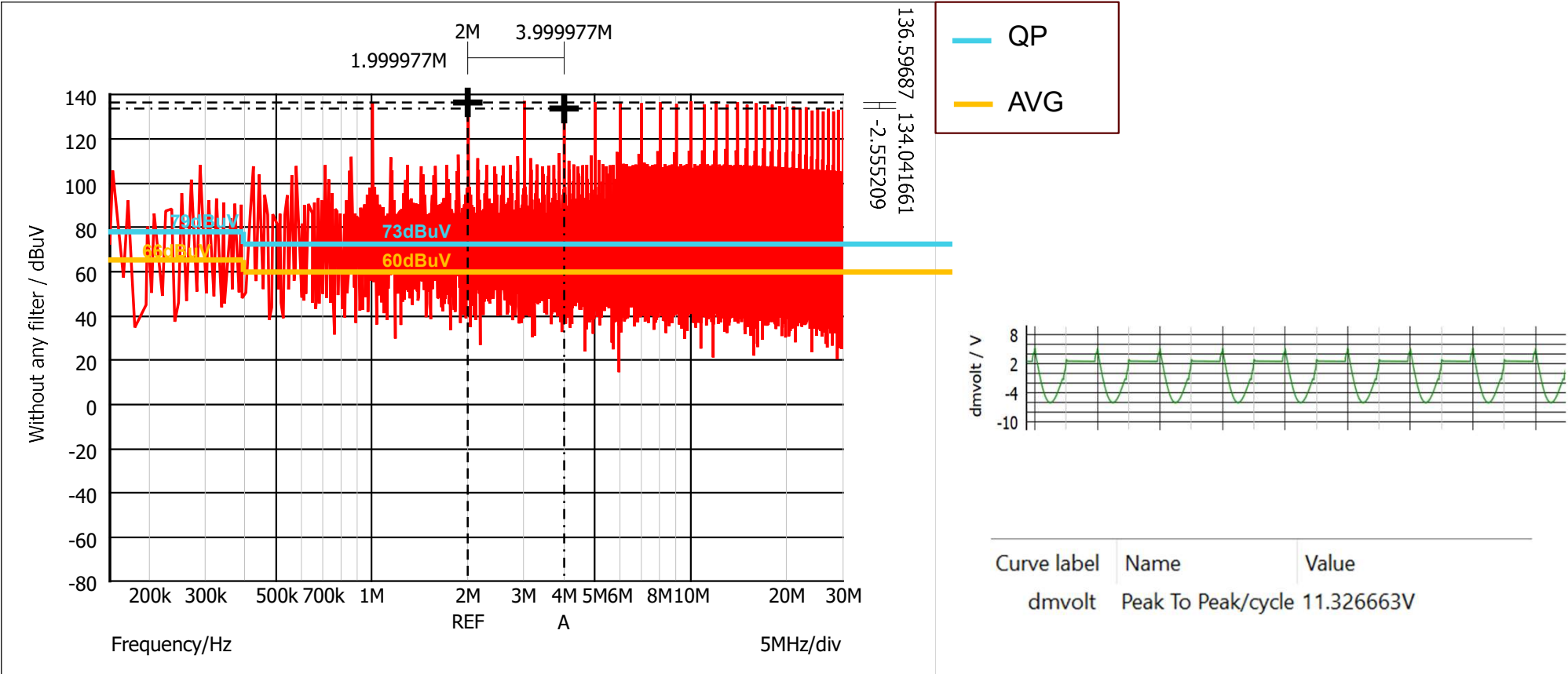
Studying the decrease in size of passives With the use of AEF

If we use passive filters only, we'll have to use larger inductors and x-capacitors for the same attenuation. Use of AEF leads to 50% decrease in size of x-capacitors and 30% decrease in inductor size in the circuit.

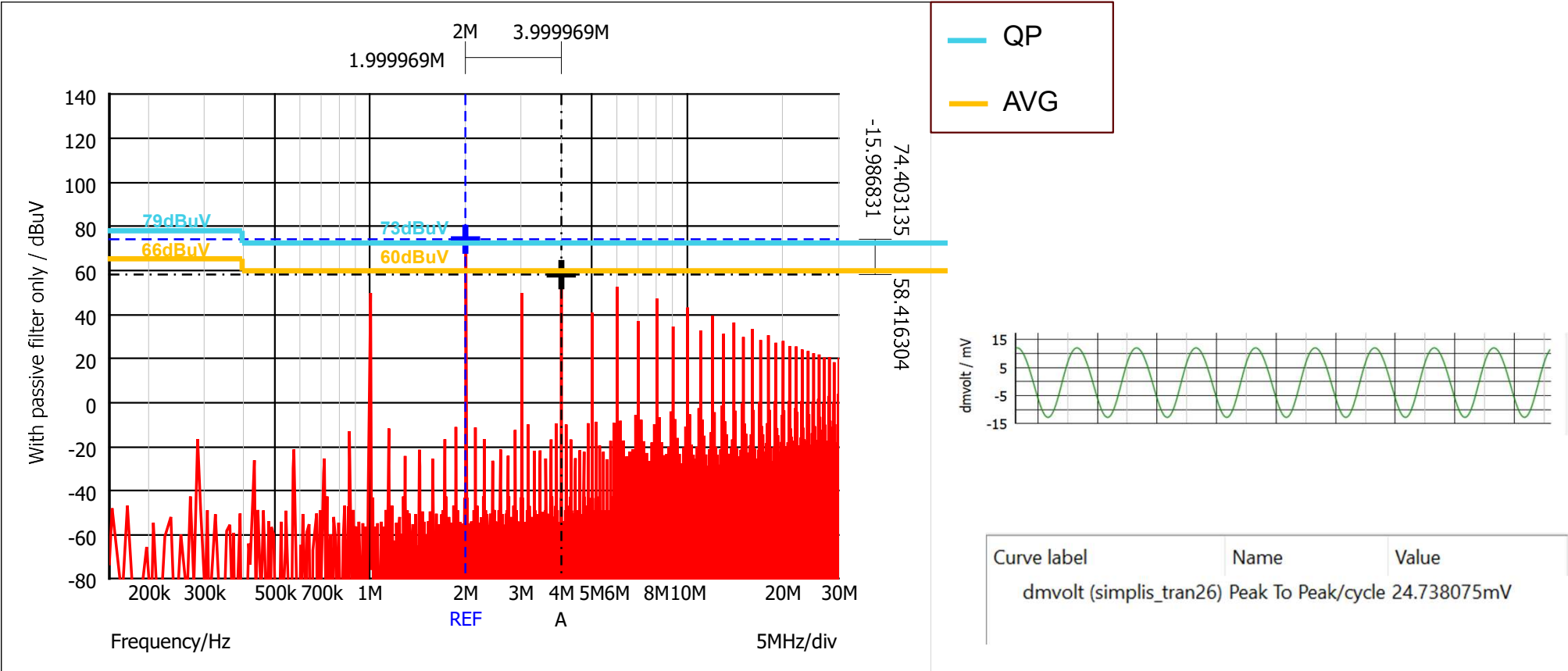
TI Information – Selective Disclosure

Configuration	Inductor size	X-capacitor size	Comments	Attenuation (in dBuV)
Only Passives (Without AEF)	2.5uH	1uF	Bulkier passives	58.87
With Passives + AEF	1.8uH (30% ↓)	500nF (50% ↓)	Lighter passives	59.33

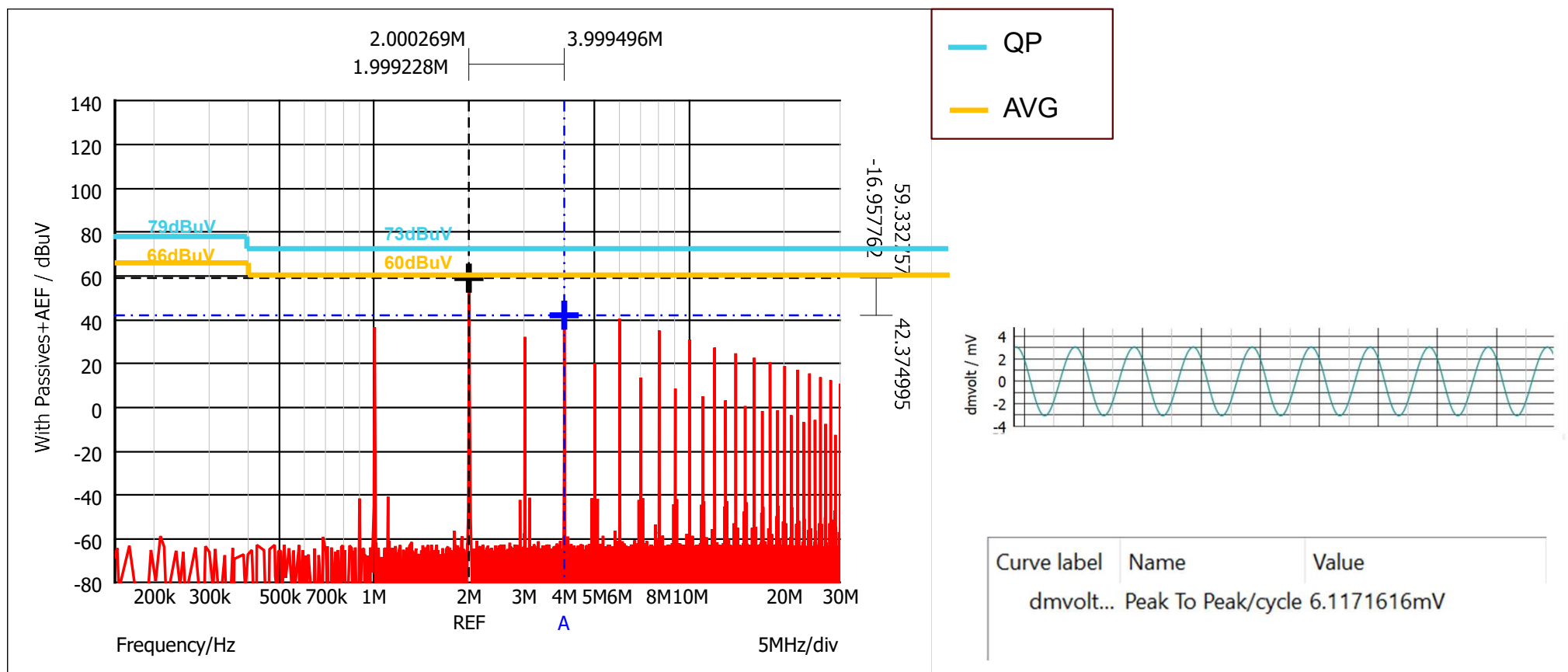
ISOP topology – DM noise profile



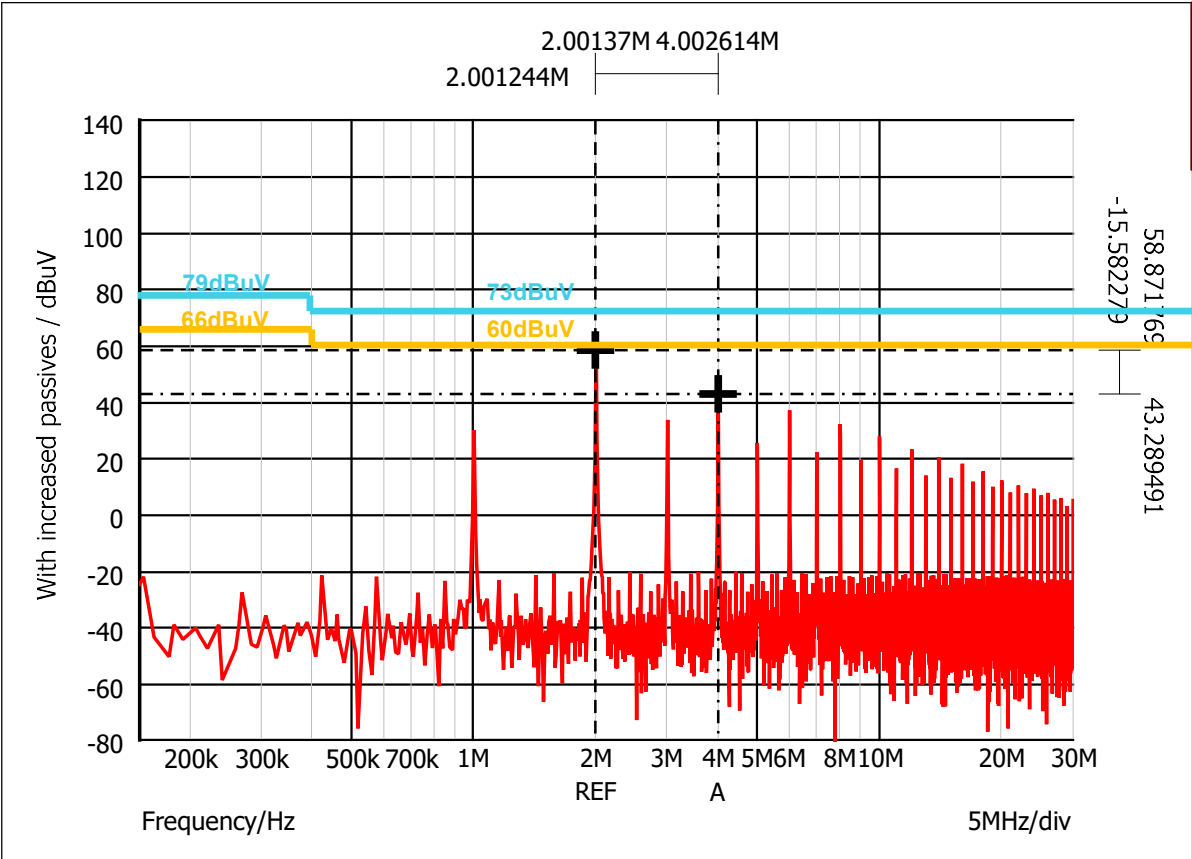
ISOP topology- DM noise profile with passives only



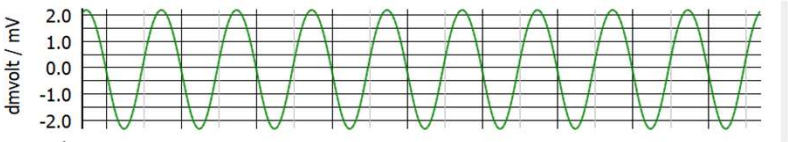
ISOP topology- DM noise profile with passives+AEF



ISOP topology- DM noise profile with increased passives



The passives were increased to 2.5uH and 1uF to get the same attenuation as the case lighter passives and AEF.

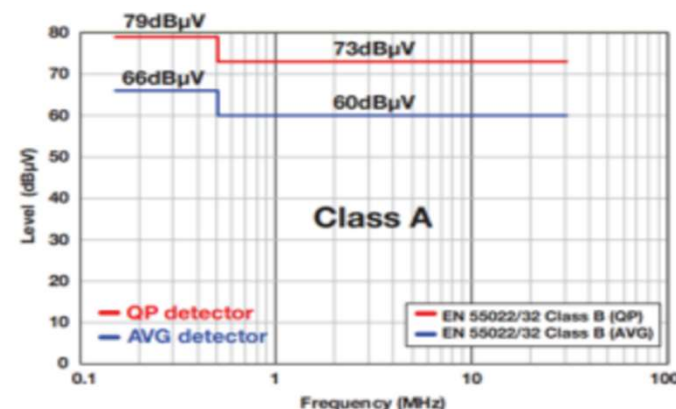


Curve label	Name	Value
dmvolt (simplis_tran7)	Peak To Peak	4.5203014mV

Results for Sentry Schematic| Summary

- Target EMI limit : CISPR 32 Class A

Sl.no.	Condition	Highest peaks level (in dBuV)	Comments
1	No filter	85.38 (1MHz peak, closely followed 10MHz peak at 80)	The spectrum from 150kHz to 30MHz was above dBuV (high noise content).
2.	With π – LC filter (500nH and 500nF) with snubber circuit	66.32 (1MHz peak only)	Low noise, below the QP limit but above the AVG limit
3.	With π – LC filter (500nH and 500nF) with snubber circuit + AEF	56.12(1MHz peak only)	Low noise safely below the QP limit and just below AVG limit too.



The EMI voltage being measured is a continuous signal. Hence its peak = quasi-peak

Thus it passes both QP and AVG detectors.

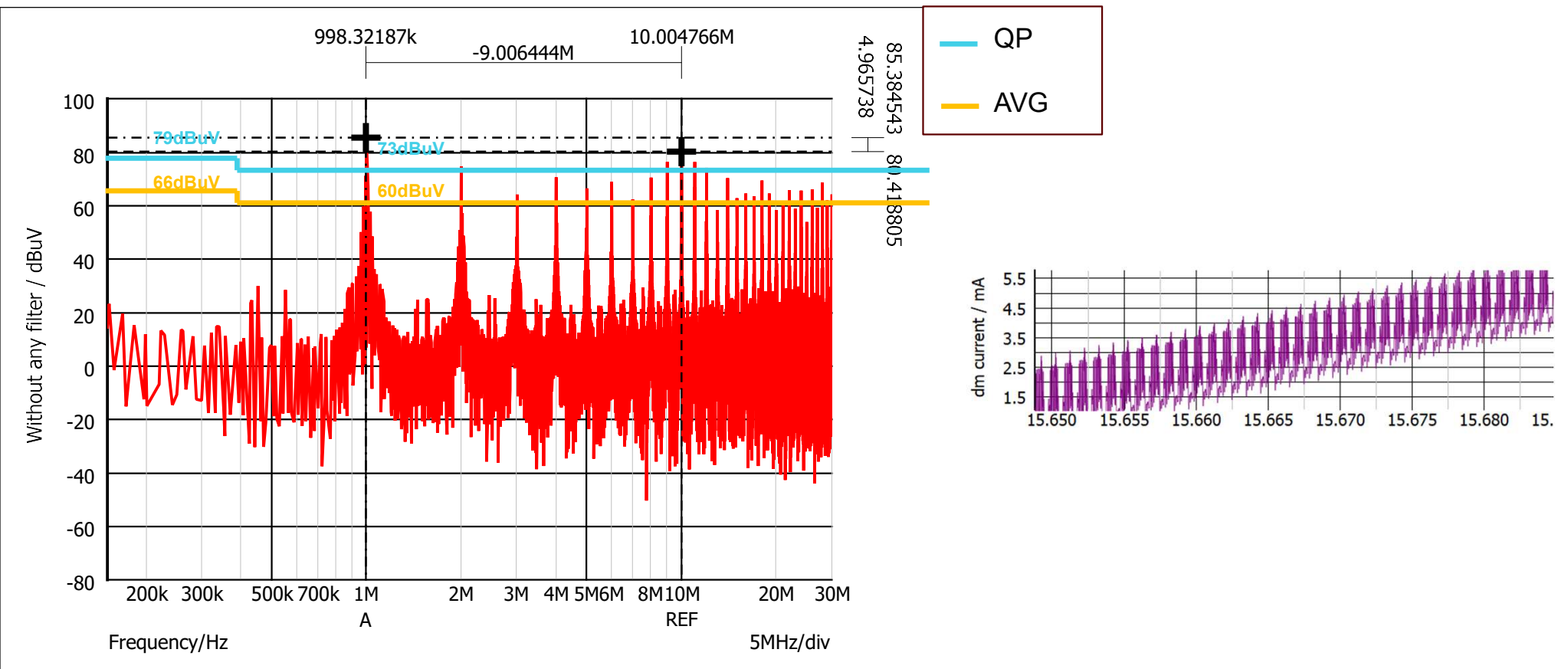
Studying the decrease in size of passives with the use of AEF

If we use passive filters only, we'll have to use larger inductors and x-capacitors for the same attenuation. Use of AEF leads to 30% decrease in size of both x-capacitors and inductor size in the circuit.

TI Information – Selective Disclosure

Configuration	Inductor size	X-capacitor size	Comments	Attenuation (in dBuV)
Only Passives (Without AEF)	800nH	800nF	Bulkier passives	55.33
With Passives + AEF	500nH (30% ↓)	500nF (30% ↓)	Lighter passives	56.12

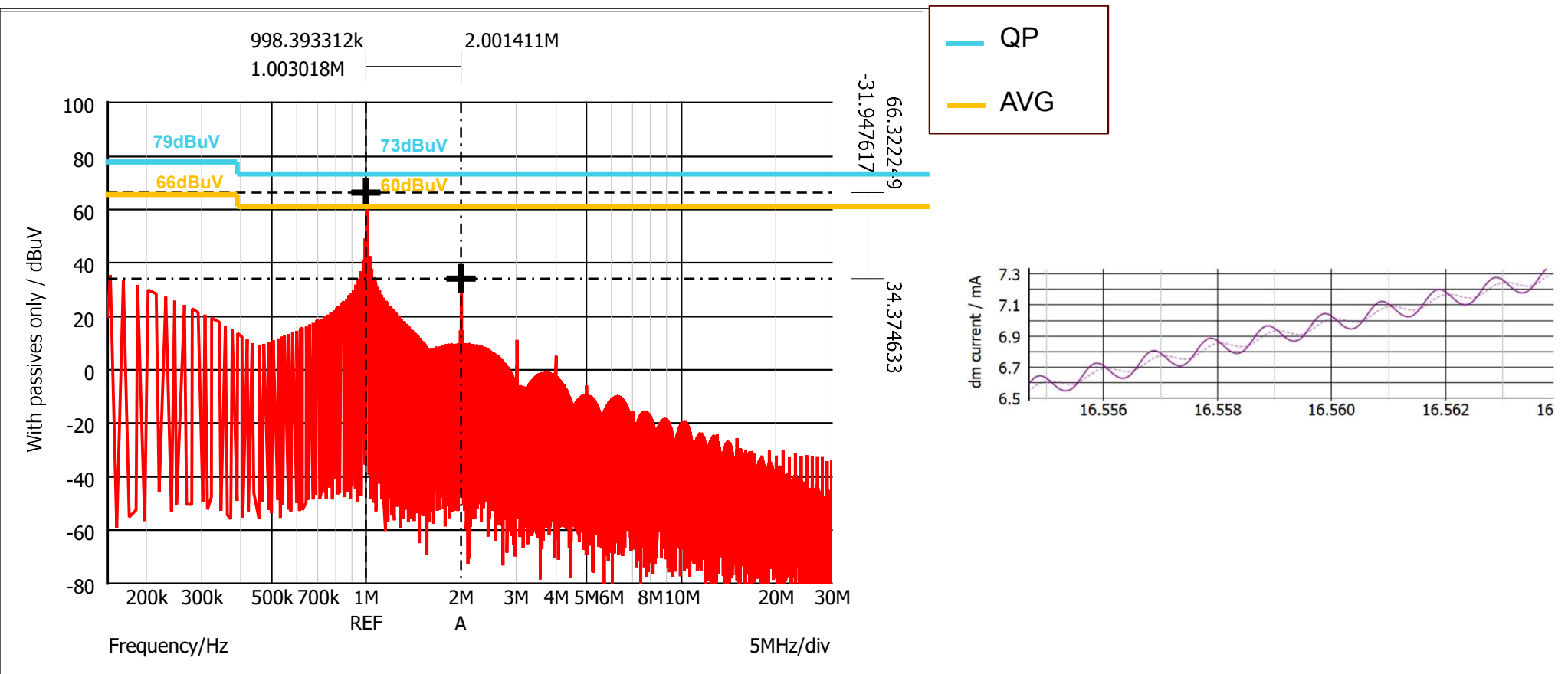
Sentry Schematic – Without any filter



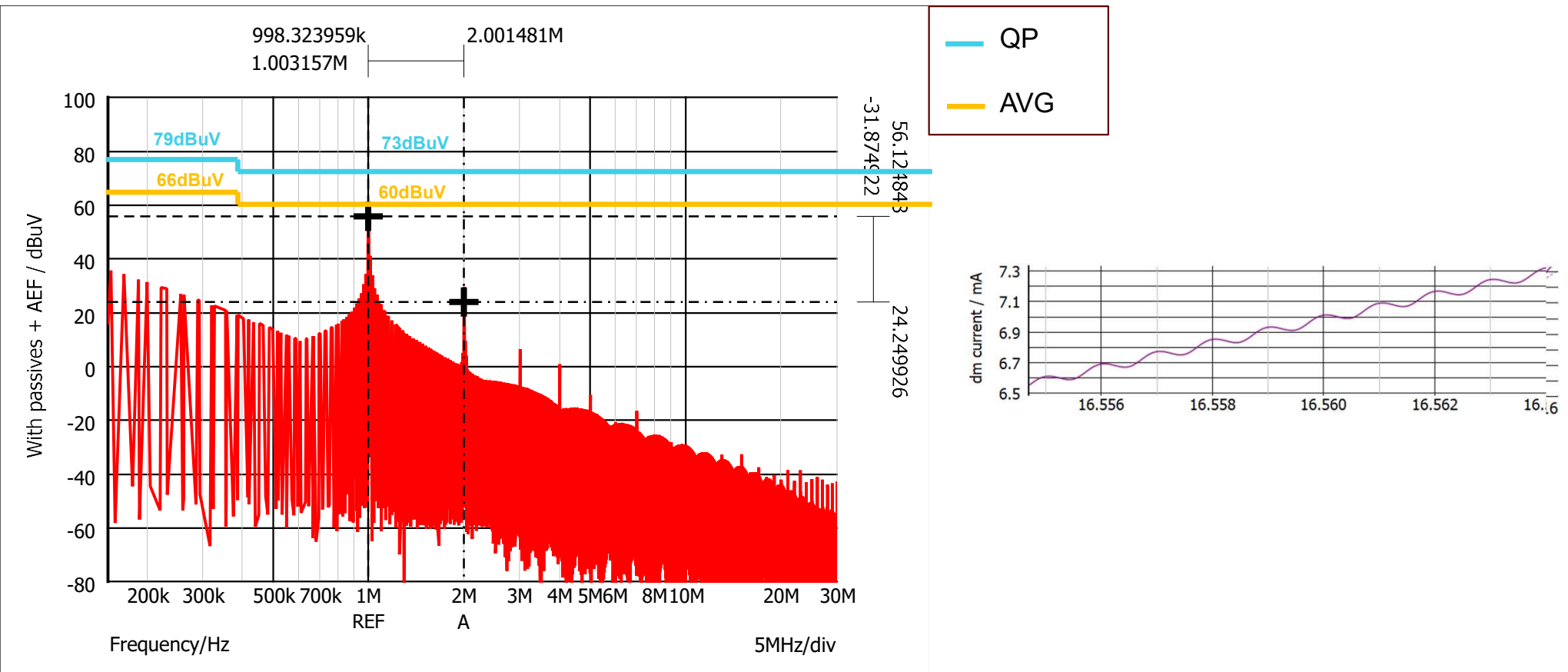
TI Information – Selective Disclosure

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Sentry Schematic – With passives only



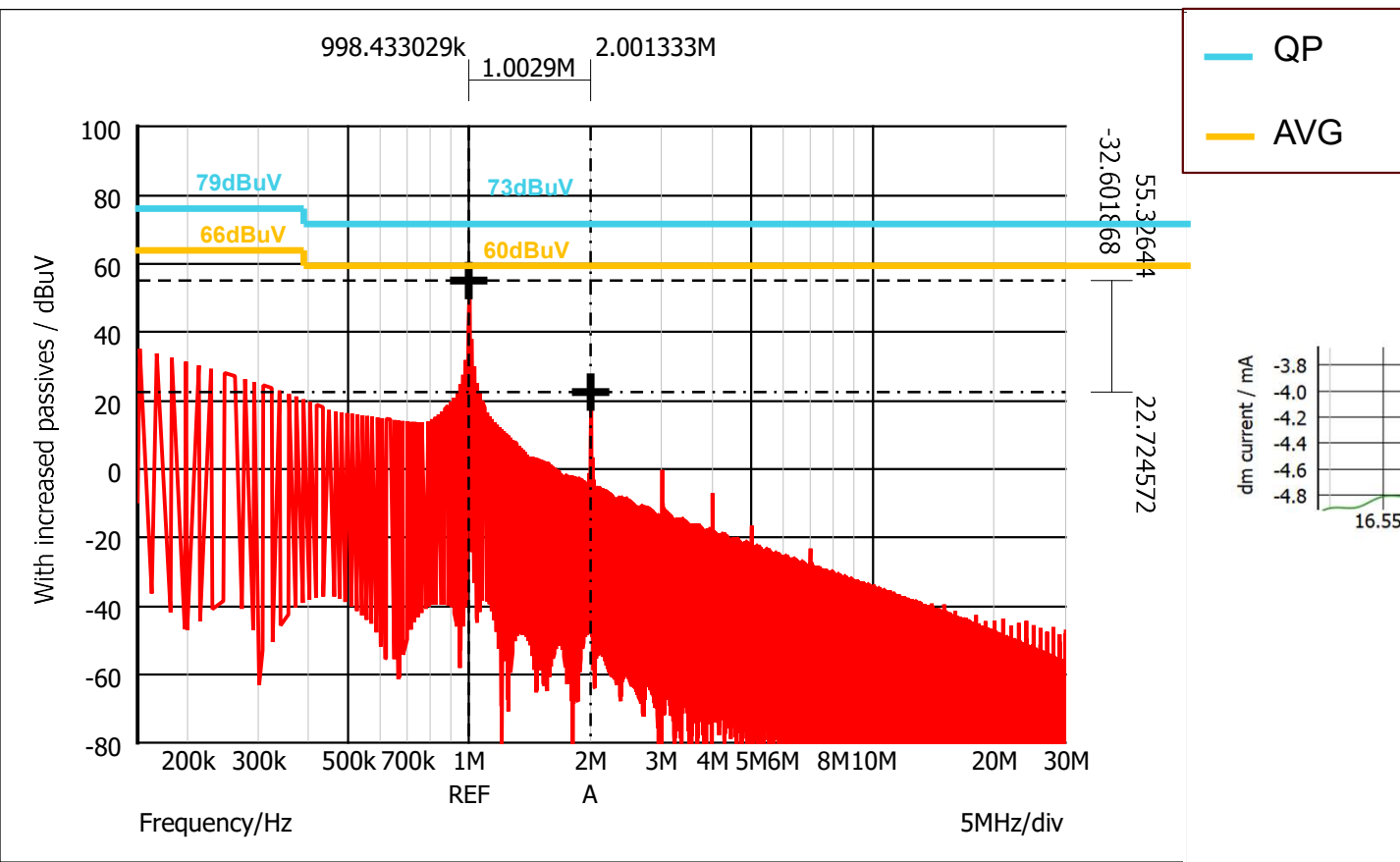
Sentry Schematic – With passives + AEF



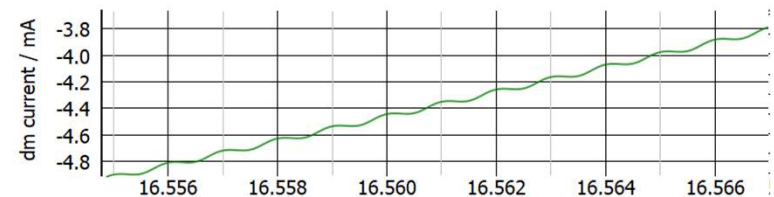
TI Information – Selective Disclosure

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Sentry Schematic – With increased passives



The passives were increased to 800nH and 800nF to get the same attenuation as the case lighter passives and AEF.



C2D (CM to DM conversion)

- The CM noise can be converted into DM noise by actively forcing the noise in the ground into the Neutral line, by using an active circuit

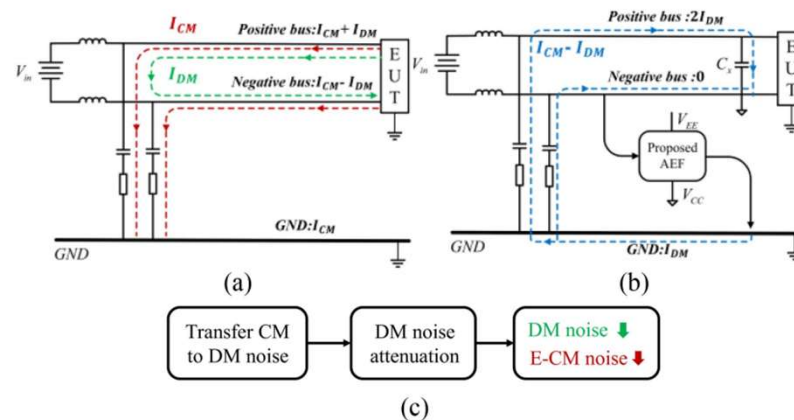
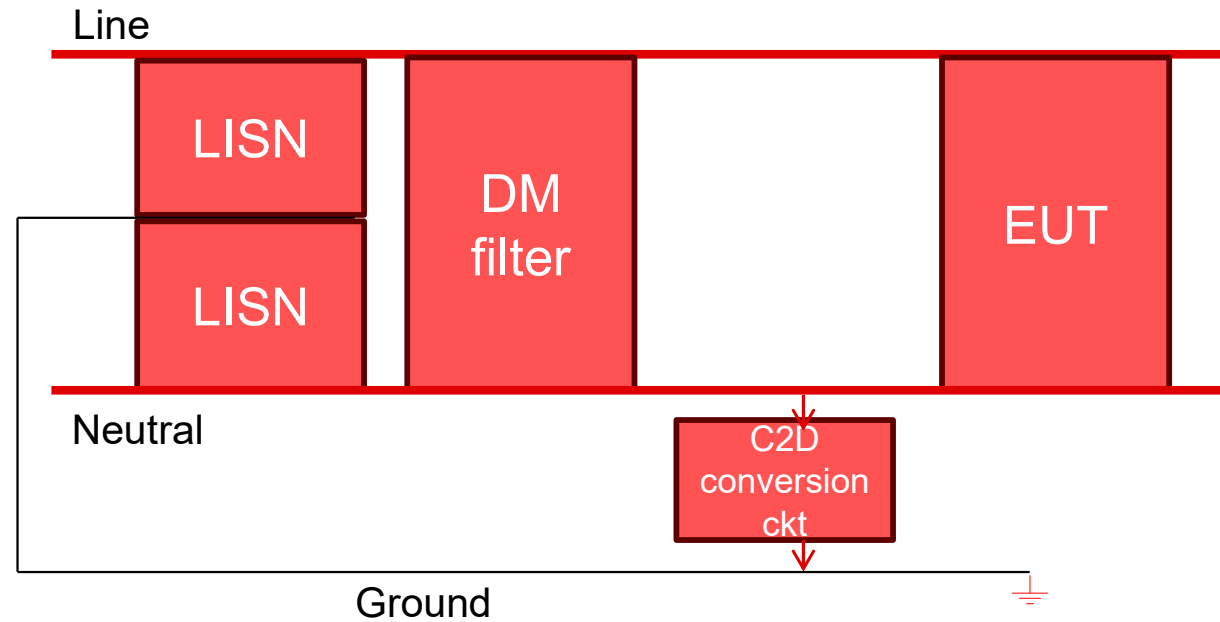
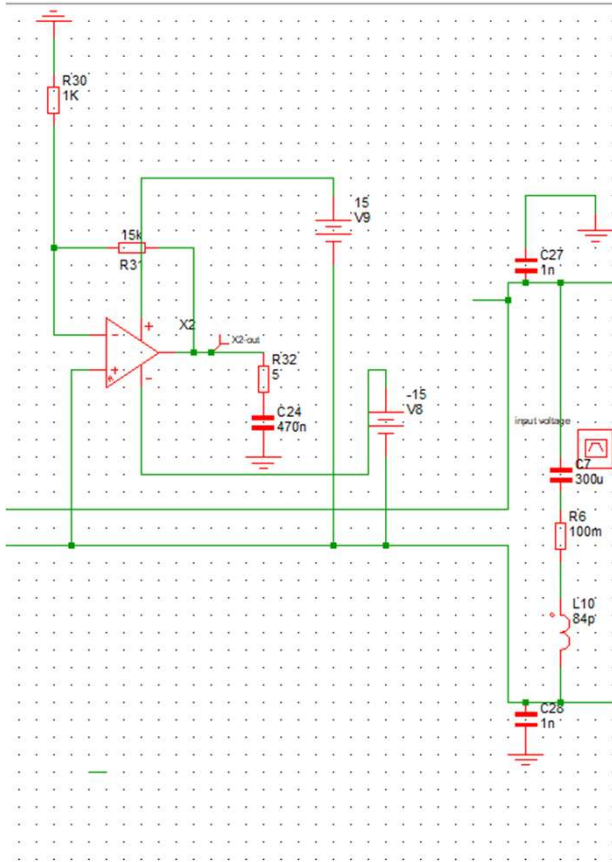


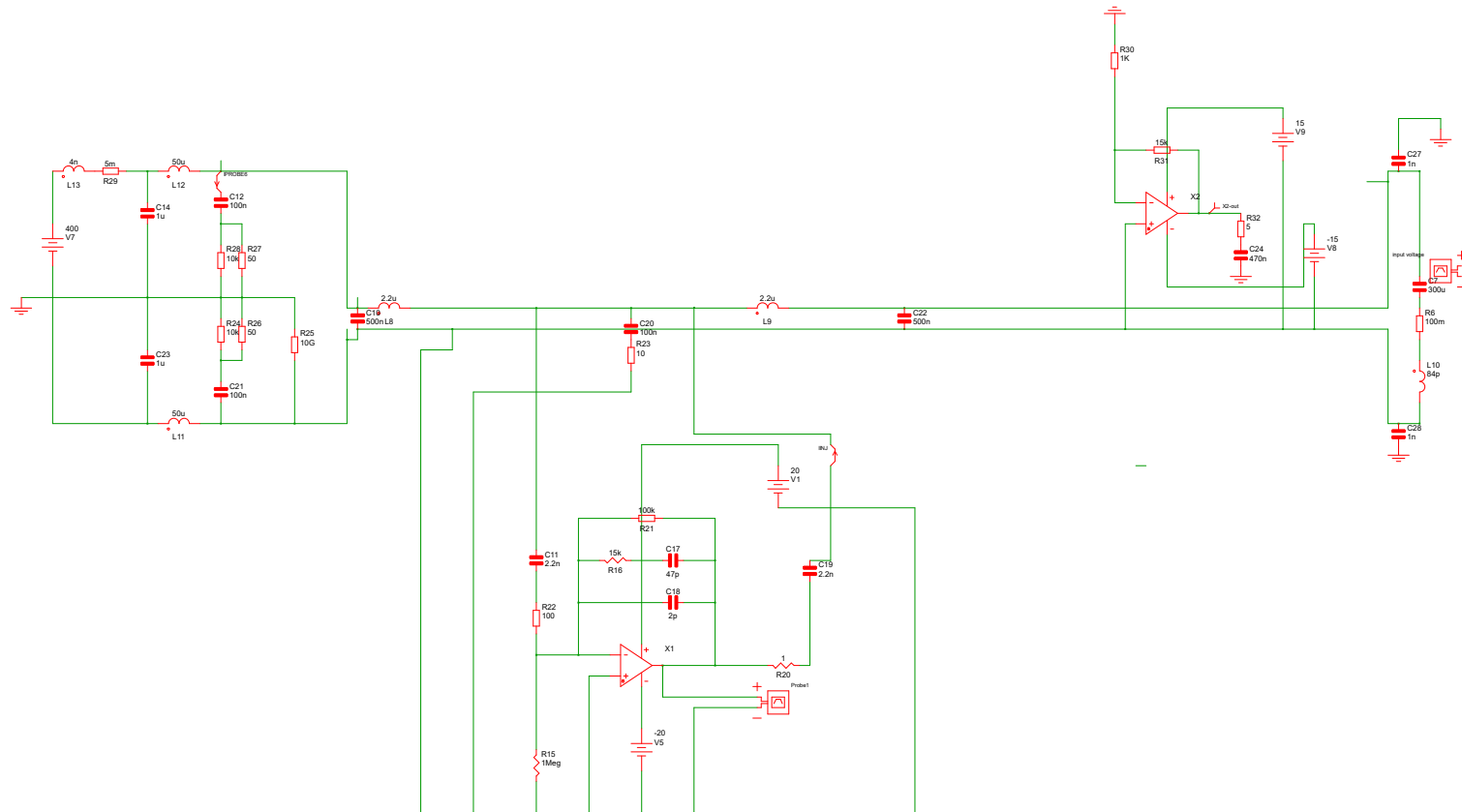
Fig. 2. (a) Noise current flow path without the AEF. (b) Compensation current flow path of the proposed C2D-AEF. (c) C2D-AEF workflow diagram.

Reference: H. Wang et al., "A Novel Active EMI Filter Based on Transferring Common-Mode Noise to Differential-Mode Noise," IEEE Transactions on Power Electronics, vol. 40, no. 7, p. 8950, July 2025.

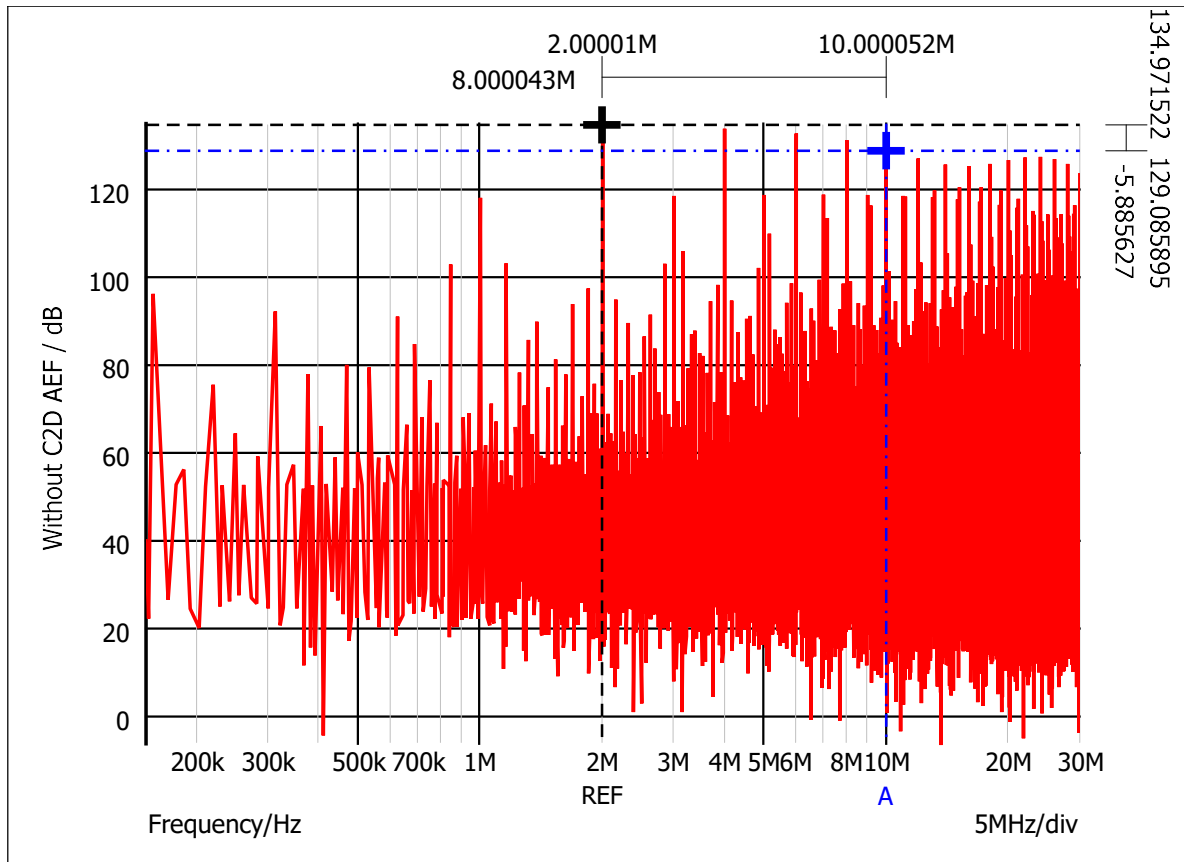
The Active circuit involved



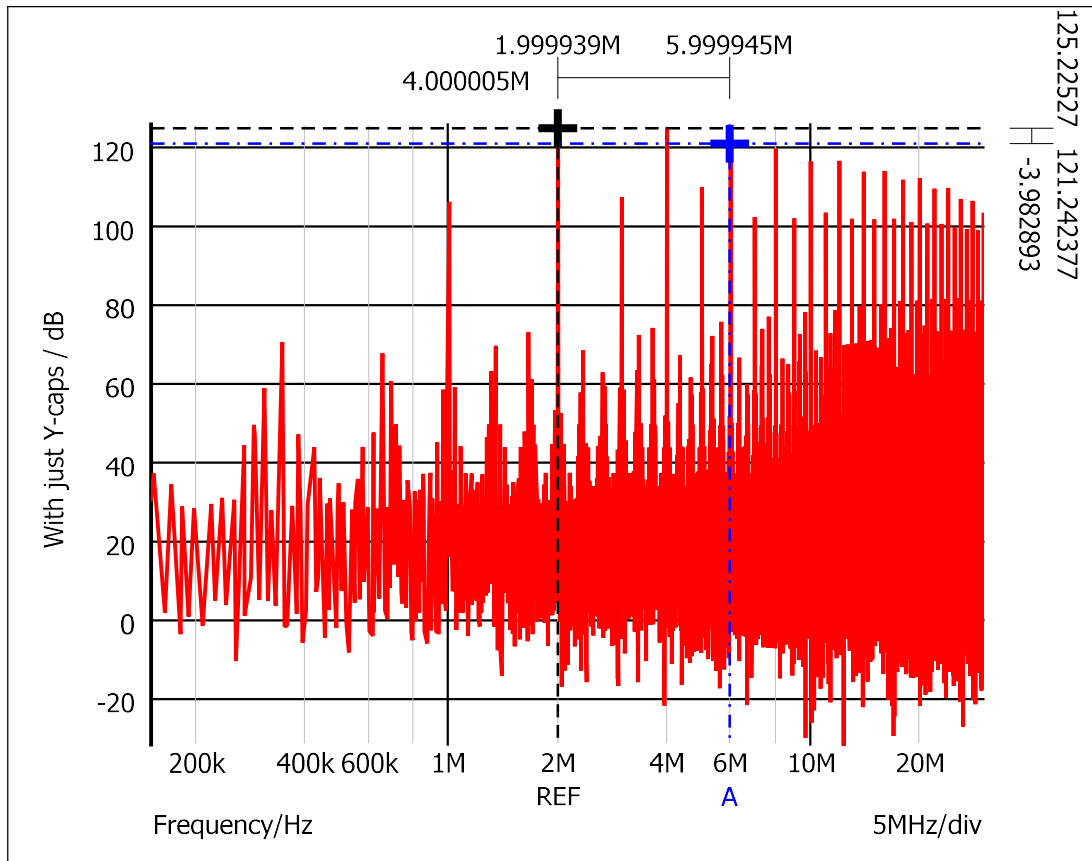
Schematic



Targeting CM using C2D | Case 1 : Without filter

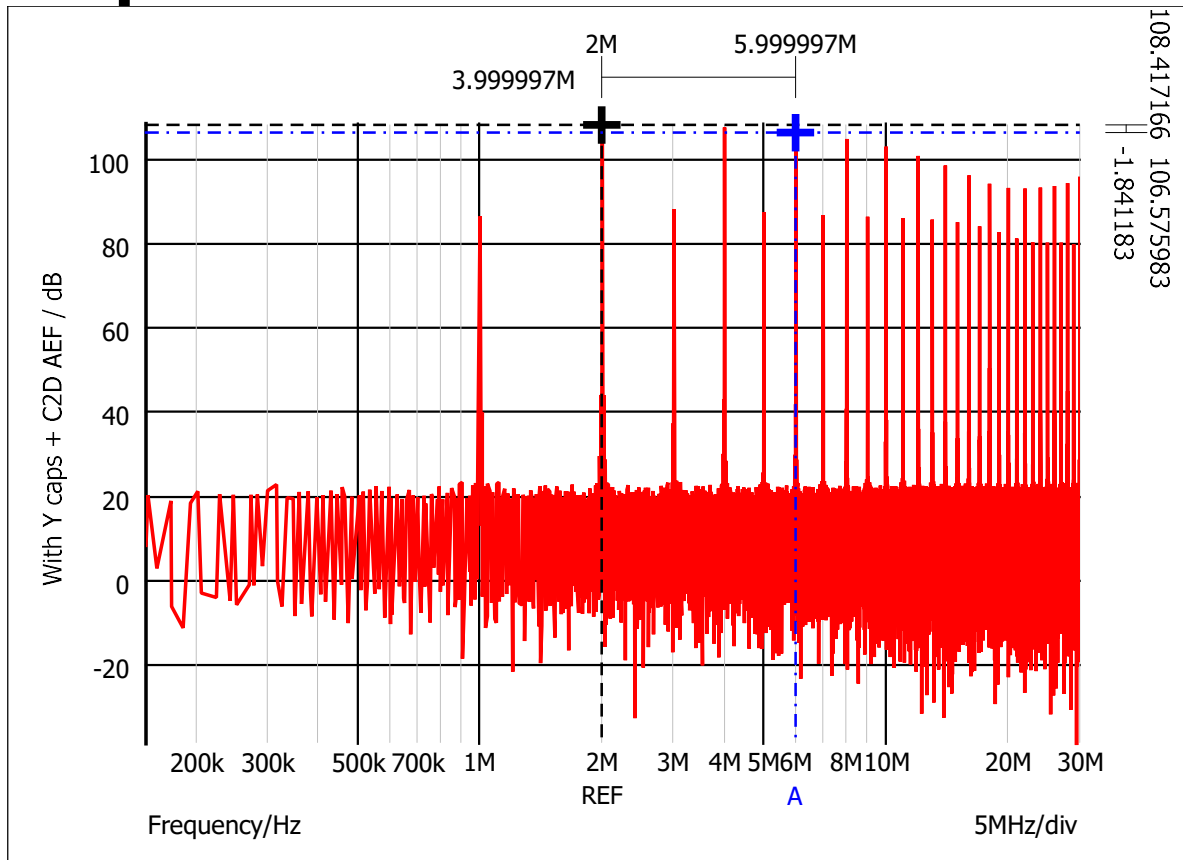


Targeting CM + DM | Case 2: With just Y-cap



-9dBuV

Targeting CM using C2D | Case 3: With C2D AEF + Y-cap



-17dBuV

THANK YOU...!